

What is Machine Learning?

A Formal Definition

- **Machine Learning** is formally defined as "the field of study that gives computers the **ability to learn without being explicitly programmed**."
 - This definition is attributed to **Arthur Samuel**, a pioneer in the field of AI and computer gaming.
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Core Concepts 🏠

- **Learning from Experience:** The core idea is that a program can improve its performance on a task by learning from data or experience.
 - **Arthur Samuel's Checkers Program (1950s):**
 - Samuel, who was not a great checkers player himself, wrote a program that learned to play checkers.
 - He had the computer play **tens of thousands of games against itself**.
 - By observing which board positions led to wins versus losses, the program learned to identify good and bad positions, eventually becoming a better player than its creator.
 - This illustrates a key principle: in general, **the more data (experience) a learning algorithm has, the better it will perform**.
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Major Types of Machine Learning

The two main categories of machine learning algorithms are:

1. **Supervised Learning:** This is the most widely used type of machine learning in real-world applications and is the focus of the first two courses in this specialization.
 2. **Unsupervised Learning:** This will be covered in the third course.
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Focus on Practical Application 🛠️

- Beyond just learning the algorithms ("the tools"), this course emphasizes the **practical skills needed to apply them effectively**.
- Knowing *how* and *when* to use different techniques is crucial for building successful machine learning systems and avoiding common pitfalls that can waste significant time and effort.